

PATTON TANG

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PROFESSIONAL EXPERIENCE

Brooks Automation | Fremont, CA

June 2026 - Present

Software Engineer

- Developed and designed C# interfaces and event-driven finite state machines to enable API control of machine elements using a publish-subscribe architecture.
- Implemented bug fixes and software features within an Agile development cycle, collaborating with senior engineers on architecture and tracking via Azure DevOps.
- Built and implemented a CSV file conversion API for clients and service engineers, transforming log data and machine configurations into a human readable format.
- Documented test cases to validate software functionality and identify potential defects, collaborating with SQA and cross-functional engineering teams.

Freelance Artist | Various, United States

May 2025 – June 2026

- Analyzed and recorded client metrics to develop data-driven recommendations aligned with client goals, evaluating branding, product diversity, inventory, and location.
- Coordinated project timelines, meetings, deliverables, and client feedback.
- Led personalized marketing strategies and custom booth design for pop-up events attracting 4,400+ attendees, generating 700%+ ROI on select product lines.

PROJECTS

TypeFighter | *React, Flask, Vercel, PostgreSQL*

- Co-deployed a typing test website to improve user words per minute via gamification methods such as through traditional score leaderboards and community driven goals.
- Designed system architecture for PostgreSQL database to be populated with typing prompts (book data) via Open Library APIs along with registered user information.
- Calculated and displayed real-time words per minute (WPM) and accuracy (%) within 10ms error with safeguards to ensure data authenticity.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, SQL, C#, Javascript, HTML, CSS
- **Frameworks:** Flask, .NET, React
- **Developer Tools:** Git, Microsoft Suite (Excel, Word, Powerpoint), Visual Studio

EDUCATION

California State University, Fullerton | Fullerton, CA

Aug 2022 - May 2025

Bachelor of Science in Computer Science

GPA: 3.5 / 4.0, cum laude

- Relevant Coursework: Web Front-End & Back-End Engineering, Algorithm Engineering, Data Structures, File Structures and Database Systems, Data Science and Big Data